LandLedger

Blockchain Property Registry & Escrow Protocol on BOT Chain

WHITEPAPER | VERSION 1.0
October 2026 · BOT Chain · Chain ID 968

PROTOCOL OVERVIEW

LandLedger is a blockchain-native property registry and escrow protocol built on BOT Chain. A government-appointed registrar registers verified land parcels on-chain with a document hash. Owners can list properties for sale; buyers pay into a smart-contract escrow; the owner confirms or rejects the offer, at which point title transfers automatically and a 1% registry fee is deducted and forwarded to the registrar's address. The result is an immutable, auditable chain of title with built-in escrow — no solicitors, no paper deeds, no settlement risk.

A technical overview of the protocol, its design, core functions, security properties, and roadmap.

1. Problem Statement

- Paper land registries — easily tampered, slow to transfer, opaque to buyers.
- Centralised digital registries — single point of failure; often inaccessible to international buyers.
- No escrow standard — buyers typically rely on bank escrow with high fees and long settlement windows.
- Duplicate registrations — same parcel sold to multiple buyers due to registry fraud.

2. Protocol Flow

```
Step
Actor
Action
```

1. Register

Registrar (Admin)

```
registerProperty(owner, title, location, size, docHash)
```

2. List

Current Owner

```
listForSale(id, priceWei) or delist(id)
```

3. Offer

Buyer

```
buy(id) — sends exact ETH price into escrow
Step
Actor
Action
```

4a. Confirm

Owner

```
confirmSale(id) — title moves, owner gets price minus 1% fee
```

4b. Reject/CancelOwner or Buyer

```
reject(id) or cancelPurchase(id) — escrow fully refunded
```

3. Contract Architecture

3.1 Data Model

Property: title, location, size, docHash, owner address, price (wei), forSale flag,

buyer address, escrow balance, registeredAt timestamp.

Transfer (history entry): owner address, price, time.

3.2 Fee Model

A configurable feeBps (basis points, default 100 = 1%, maximum 500 = 5%) is deducted from the sale price on confirmation and forwarded to the admin / registrar address. This creates a self-sustaining registry funded by property transactions.

```
Parameter
```

Default

Maximum

`Registry Fee (feeBps)`
100 bps (1%)

500 bps (5%)

`Fee Recipient`
admin / registrar

—

`Escrow held by`
Smart contract

—

3.3 Write Functions

`Function`
`Role`
Description

`registerProperty(...)`
Admin

Registers a verified land parcel

`listForSale(id, price)`
Owner

Lists property at a fixed price

`delist(id)`
Owner

Removes listing (no active offer)

`buy(id)`
Buyer

Places offer — sends exact ETH into escrow

`confirmSale(id)`
Owner

Accepts offer — transfers title, releases escrow

`reject(id)`
Owner

Rejects offer — refunds buyer

`cancelPurchase(id)`
Buyer

Cancels own offer — refunded

setFee(bps)

Admin

Adjusts registry fee (max 5%)

`transferAdmin(addr)`
Admin

Transfers admin/registrar role

3.4 Transfer History

Every sale appends a Transfer record to history[id] — an immutable, on-chain chain of title that anyone

can audit.

4. Security Properties

4.1 Escrow Safety

Funds are held by the contract, not the owner. The owner cannot access the escrow until confirmSale is called, after which the transfer is atomic.

4.2 Single-Offer Constraint

require(p.buyer == address(0)) prevents a second buyer from depositing while a prior offer is open, eliminating double-escrow risk.

4.3 Atomic Title Transfer

p.owner = buyer executes in the same transaction as the ETH transfer, so title and payment are inseparable.

4.4 Reentrancy

Escrow state is zeroed (p.escrow = 0, p.buyer = address(0)) before _send() in both reject and confirmSale paths.

5. Events

Event	Parameters
Registered	id, owner
Listed	id, price
Delisted	id
OfferMade	id, buyer, amount
OfferCancelled	id, buyer
Sold	id, from, to, price

6. Deployment

Parameter	Value
Chain	BOT Chain (Chain ID 968)
RPC	https://rpc.botchain.ai
Explorer	

https://scan.botchain.ai

`Deployer`
Becomes admin / registrar

`Default fee`
1% (100 bps)

7. Legal Notice

LandLedger automates the technical transfer of on-chain title records. It does not replace legal land registration requirements in your jurisdiction. Physical property rights are governed by local law; parties must ensure all statutory requirements (stamp duty, survey, planning permissions) are satisfied separately.

Not audited.

8. Roadmap

`Milestone`
Description

v1.1

Multi-signature registrar (2-of-3 admin approval)

v1.2

ERC-20 stablecoin payments (USDC)

v1.3

Dispute resolution module with arbitration address

v2.0

NFT-based title deeds (ERC-721 with transfer restrictions)